METRIC

DOD-C-70513A (AR)
30 July 1991
SUPERSEDING
DOD-C-70513 (AR)
23 August 1985

MILITARY SPECIFICATION

CARTRIDGE, 120 mm, KE SLUG

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. - This specification establishes the requirements and quality assurance provisions for manufacturing, assembling, loading and packing two types of cartridges designated as:

Cartridge, Slug, 120 mm, Cannon Proof
Cartridge, Slug, 120 mm, Case Base Test

2.0 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. - Unless otherwise specified, the following specifications, standards, and handbooks of the issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-70451	Primer, Electric, M125 Load, Assemble, Pack

Beneficial comments (recommendations, additions, deletions) and any other pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, Attn: AMSMS-QAA(D), Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-1315

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DOD-C-63485	Case Base and Seal Assembly
DOD-P-70639	Propellant, JA-2, for use in Cartridge, 120 mm, KE Cannon Proof Slug
DOD-P-64009	Projectile, 120 mm TPCSDS-T, M865 Metal Parts for
DOD-P-64035	Propellant, JA2, for use in Cartridge, 120 mm, APFSDS-T, M829

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment
MIL-STD-1167	Ammunition Data Card
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. - The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12524865	Cartridge, Slug 120 mm, Cannon Proof
12525027	Cartridge, 120 mm, Slug, Case Base Test

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS

Title 49	Transportation, Parts 100-199, Paragraph 173.54

ARMY REGULATIONS

AR-700-15	Packing of Material

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D624	Rubber Property - Tear Resistance, Test for
ASTM D789	Nylon Injection Molding and Extrusion Materials, Specification for
ASTM D790	Flexural Properties of Plastics and Electrical Insulating Materials, Tests for
ASTM D792	Specific Gravity and Density of Plastics by Displacement, Tests for
ASTM D2240	Rubber Property - Durometer Hardness, Test for
ASTM D2632	Rubber Property - Resilience by Vertical Rebound, Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.2 Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3.0 REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing 12524865 or 12525027 and the drawings and specifications referenced therein.

3.3 Propellant weight. - The weight of the propellant shall not differ by more than plus or minus 4 grams from the selected weight for the particular lot of propellant used and cartridge being loaded.

3.3.1 Cannon proof cartridge. - The propellant weight for the Cannon Proof Slug, P/N 12524865 as delivered, shall be defined by the procedure in 6.7. The propellant weight may be adjusted as required at the proving ground to meet the required chamber pressure.

3.3.2 Case base test cartridge. - The propellant weight of the Case Base Test Slug, P/N 12525027 as delivered, shall be as required to meet the provisions in paragraph 3.4.1.2. The propellant weight may be adjusted as required at the proving ground to meet the required chamber pressure.

3.4 Performance. - The cartridge shall comply with the following requirements.

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3.4.1 Chamber pressure.

3.4.1.1 Cannon proof cartridge. - The chamber pressure for the Cannon Proof Slug, P/N 12524865, as delivered shall be within the range of 6700 to 7070 bars when fired at +63 °C as measured 8.89 + 1.3 cm forward of the breech.

3.4.1.2 Case base test cartridge. - The chamber pressure for the Case Base Test Slug, P/N 12525027, shall be defined in DOD-C-63485.

3.4.1.3 Primer test cartridge. - The chamber pressure for the Primer Test Slug shall be defined in MIL-P-70451.

3.4.4 Misfire. - The cartridge shall not fail to fire from the weapon.

3.4.5 Metal parts security. - There shall be no indication of parts failure in the gun bore.

3.5 Adhesive bond. - The test sample cartridge case/cap case shall separate at other than the glue joint interface under a minimum tensile load as defined on the cartridge drawing.

3.6 Primer electrical resistance. - The primer electrical resistance shall not exceed the following:

• Primer electrode to primer body	1.8 ohms maximum, 0.8 ohms min.
Primer body to case base	0.1 ohms maximum

3.7 Primer deterioration. - See paragraph 6.6.

3.8 Workmanship. - The cartridge and all components shall comply with the provision of MIL-A-2550. The cleaning method shall not be injurious to any of the parts; nor shall the parts be contaminated by the cleaning agents.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection

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4.3 Quality conformance inspection.

4.3.1 Lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain a maximum of:

- One lot of propellant
- Combustible cases from one lot interfix number from one manufacturer
- Primers from one lot interfix number from one manufacturer
- Case base and seal assemblies from one lot interfix number from one manufacturer
- Projectile metal parts from one lot interfix number from one manufacturer

4.3.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the classification of defects, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be used if approved by the Government.

b. Interface characteristics. - Interface characteristics identified by the interface symbol on the drawings are identified by an "I" preceding the defect number in the classification of defects and tests.

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.1	TITLE Counterweight, Slug, Cannon Proof		SHEET 1 OF 1		DRAWING NUMBER 12524858
					NEXT HIGHER ASSEMBLY 12524861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter undersize		0.40%	3.2	Gage
Minor					
201	Pitch diameter oversize		0.65%	3.2	Gage
202	Weight not correct		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.2	Counterweight, Slug, Case Base Test				12525024
					NEXT HIGHER ASSEMBLY 12525026
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Marking missing or incorrect		100%	3.2	Visual
Major					
101	Pitch diameter undersize		0.40%	3.2	Gage
Minor					
201	Pitch diameter oversize		0.65%	3.2	Gage
202	Weight not correct		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.3	TITLE Spring, Disc		SHEET 1 OF 1		DRAWING NUMBER 12526678
					NEXT HIGHER ASSEMBLY 12524865, 12525027
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Hardness	1/		3.2	Gage
204	Height of formed part		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.8	Visual
206	Absence of post-plating bake (hydrogen embrittlement relief)	2/ 3/		3.2	Certification

NOTES:
1/ Three pieces per heat treat lot, no defects allowed.
2/ Each lot.
3/ Applies to electrodeposited zinc plated parts only.

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.4	Bushing				12524925
					NEXT HIGHER ASSEMBLY
					12524861, 12525026
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Smallest outside diameter		0.65%	3.2	Gage
202	Distance from front face to shoulder		0.65%	3.2	Gage
203	Protective finish missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.5	Screw, Cylindrical				1224928
					NEXT HIGHER ASSEMBLY
					12524861, 12525026
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pitch diameter of threads		0.65%	3.2	Gage
202	Distance from front end of threads to shoulder		0.65%	3.2	Gage
203	Protective finish missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.6	TITLE Rubber, Silicone		SHEET 1 OF 1		DRAWING NUMBER 12524837
					NEXT HIGHER ASSEMBLY 12524862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tear strength	2		3.2	4.4.3.1.1
102	Hardness	2		3.2	4.4.3.1.1
Minor					
201	Density	2		3.2	4.4.3.1.1
202	Tensile strength	2		3.2	4.4.3.1.1
203	Elongation	2		3.2	4.4.3.1.1
204	Resilience	2		3.2	4.4.3.1.1

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.7	TITLE Band and Ring, Holding Material		SHEET 1 OF 1		DRAWING NUMBER 12524913
					NEXT HIGHER ASSEMBLY 12524862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Specific gravity	2		3.1	4.4.3.1.2
102	Moisture content, maximum	2		3.1	4.4.3.1.2
103	Flexural modulus of elasticity, minimum	2		3.1	4.4.3.1.2
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.8	TITLE Body, Slug		SHEET 1 OF 2		DRAWING NUMBER 12524863
					NEXT HIGHER ASSEMBLY 12524862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Aft bourrelet diameter		0.40%	3.2	Gage
102	Forward bourrelet diameter		0.40%	3.2	Gage
Minor					
201	Length from forward end to rear of aft bourrelet		0.65%	3.2	Gage
202	Length of aft bourrelet		0.65%	3.2	Gage
203	Presence of knurl, aft band seat		0.65%	3.2	Visual
204	Length of forward bourrelet		0.65%	3.2	Gage
205	Second largest diameter of aft bourrelet		0.65%	3.2	Gage
206	Presence of grooves in forward bourrelet		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.8	TITLE Body, Slug		SHEET 2 OF 2		DRAWING NUMBER 12524863
					NEXT HIGHER ASSEMBLY 12524862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Presence of chamfer on forward end of forward bourrelet		0.65%	3.2	Visual
208	Proper material		100%	3.2	Certification
209	Proper protective finish		100%	3.2	Certification
210	Threads damaged or rolled over		0.65%	3.2	Visual
211	Damage to protective finish		0.65%	3.2	Visual
212	Forward thread size		0.65%	3.2	Gage
213	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.9	TITLE Body, Slug Assembly		SHEET 1 OF 3		DRAWING NUMBER 12524862
					NEXT HIGHER ASSEMBLY 1524861, 12525026
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Adhesive strength of sealing cuff		100%	3.2	4.4.3.1.1
102	Band diameters	4.4.3.1.2		3.2	Gage
103	Presence of grooves in band		0.40%	3.2	Visual
104	Body must not be heated above 155 °C during curing of sealing cuff		100%	3.2	Certification
105	Sealing cuff diameter	4.4.3.1.1		3.2	Gage
106	Shore hardness of sealing cuff		0.40%	3.2	4.4.3.1.1
107	Improper post cure of sealing cuff		100%	3.2	Certification
108	Verify proper material for holding ring, sealing cuff and band		100%	3.2	Certification
109	Forward bourrelet diameter, maximum		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.9	TITLE Body, Slug Assembly		SHEET 2 OF 3		DRAWING NUMBER 12524862
					NEXT HIGHER ASSEMBLY 1524861, 12525026
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of counterbore at M8 threads		0.65%	3.2	Gage
202	Length from rear of aft bourrelet diameter to aft edge of sealing cuff	4.4.3.1.1		3.2	Gage
203	Length of sealing cuff outside diameter	4.4.3.1.1		3.2	Gage
204	Depth of counterbore of mounting holes		0.65%	3.2	Gage
205	Pitch diameter, mounting holes threads		0.65%	3.2	Gage
206	Length of M8 threads from surface of sealing cuff, minimum		0.65%	3.2	Gage
207	Depth of drilled hole for M8 thread from surface of sealing cuff		0.65%	3.2	Gage
208	True position of mounting holes		0.65%	3.2	Gage
NOTES:					

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.3.2.9	Body, Slug Assembly				12524862
					NEXT HIGHER ASSEMBLY
					1524861, 12525026
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
209	Lot identification of sealing cuff missing or incomplete		0.65%	3.2	Visual
210	Identification markings missing or incomplete		0.65%	3.2	Visual
211	Sealing cuff damaged or porous		0.65%	3.2	Visual
212	Weight not correct		0.65%	3.2	Gage
213	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.10	Projectile, Cannon Proof Slug				12524861
					NEXT HIGHER ASSEMBLY
					12524865
CATEGORY	EXAMINATION OR TEST	NO.OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cylindrical screw torque not correct	1/		3.2	4.4.2.3
102	Adhesive missing from fin or counterweight threads	2/		3.2	Visual 3/
103	Counterweight torque not correct	1/		3.2	4.4.2.3
104	Missing adhesive on inner ring		0.40%	3.2	Visual 3/
105	Inner ring does not cover screw access holes		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
1/ Two samples with zero defects per shift
2/ Four samples with zero defects per shift
3/ Verify prior to assembly

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.11	TITLE Projectile, Slug, Base Case Test		SHEET 1 OF 1		DRAWING NUMBER 12525026
					NEXT HIGHER ASSEMBLY 12525027
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cylindrical screw torque not correct	1/		3.2	4.4.2.3
102	Adhesive missing from fin or counterweight threads	2/		3.2	Visual 3/
103	Counterweight torque not correct	1/		3.2	4.4.2.3
104	Missing adhesive on inner ring		0.40%	3.2	Visual 3/
105	Inner ring does not cover screw access holes		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
1/ Two samples with zero defects per shift
2/ Four samples with zero defects per shift
3/ Verify prior to assembly

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.12	TITLE Cartridge, 120 mm, Cannon Proof Slug		SHEET 1 OF 3		DRAWING NUMBER 12524865
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer holes (or plug in flash tube) missing		100%	3.2	Visual
Major					
101	Adhesive joint incomplete or not smoothly contoured		100%	3.2	Visual
102	Propellant charge weight not correct		100%	3.3	4.4.1
103	Minimum strength requirements not met on adhesive joint	4.4.2.1		3.2	4.4.2.1
104	Cartridge case base coat thickness not correct	4.4.2.2		3.2	4.4.2.2
105	Cartridge case finish coat thickness not correct	4.4.2.2		3.2	4.4.2.2
106	Missing or damaged primer "O" ring		100%	3.2	Visual
107	Primer torque not correct	1/		3.2	4.4.2.3

NOTES: 1/ Two samples with zero defects per shift

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.3.2.12	Cartridge, 120 mm, Cannon Proof Slug				12524865
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Primer depth below surface of case base not correct		100%	3.2	Gage
I 109	Cartridge does not chamber		100%	3.2	4.4.4
110	Primer circuit resistance not correct (prior to assembly into cartridge)		100%	3.6	4.4.5
111	Resistance, primer base to case base not correct		100%	3.6	4.4.5
112	Evidence of separation or peeling of case coating		100%	3.2	4.2.2.1
113	Retaining ring not seated in case base groove		100%	3.2	Gage
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.12	TITLE Cartridge, 120 mm, Cannon Proof Slug		SHEET 3 OF 3		DRAWING NUMBER 12524865
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Performance Test				
	63 °C (Expendable Tube)	5			
Critical	None defined				
Major					
101	Chamber pressure			3.4.1	4.4.6.3.1
102	Misfire			3.4.4	4.4.6.3.2
103	Metal parts security			3.4.5	4.4.6.3.3
104	Case base extraction			3.4.6	4.4.6.3.4

NOTES:

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.13	TITLE Cartridge Slug, 120 mm, Case Base Test		SHEET 1 OF 2		DRAWING NUMBER 12525027
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer holes (or plug in flash tube) missing		100%	3.2	Visual
Major					
101	Adhesive joint incomplete or not smoothly contoured		100%	3.2	Visual
102	Propellant charge weight not correct		100%	3.3	4.4.1
103	Minimum strength requirements not met on adhesive joint	4.4.2.1		3.2	4.4.2.1
104	Cartridge case base coat thickness not correct	4.4.2.2		3.2	4.4.2.2
105	Cartridge case finish coat thickness not correct	4.4.2.2		3.2	4.4.2.2
106	Missing or damaged primer "O" ring		100%	3.2	Visual
107	Primer torque not correct	1/		3.2	4.4.2.3

NOTES: 1/ Two samples with zero defects per shift

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.13	TITLE Cartridge Slug, 120 mm, Case Base Test		SHEET 2 OF 2		DRAWING NUMBER 12525027
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Primer depth below surface of case base not correct		100%	3.2	Gage
109	Cartridge does not chamber		100%	3.2	4.4.4
110	Primer circuit resistance not correct (prior to assembly into cartridge)		100%	3.6	4.4.5
111	Resistance, primer base to case base not correct		100%	3.6	4.4.5
112	Evidence of separation or peeling of case coating		100%	3.2	4.2.2.1
113	Retaining ring not seated in case base groove		100%	3.2	Gage
114	Diameter of glue joint, maximum		100%	3.2	Gage
Minor					
201	Cartridge marking incorrect, missing or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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4.3.3 Testing. - Testing is described in the Quality Conformance Inspection tables.

4.3.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph/Reference/Inspection Method" column in the tables starting with Paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4 Methods of inspection.

4.4.1 Propellant weight. - The propellant shall be weighed and check weighed at 100%. Check weighing shall be a completely independent operation from the original weighing using a different scale and, if performed manually, by a different operator. Any propellant charge that does not meet the applicable weight requirement shall be classified defective and removed from the lot.

4.4.2 Cartridge case.

4.4.2.1 Adhesive joint strength test. - The integrity of the adhesive joint shall be verified. The selected case cap and cartridge case shall be attached to the appropriate test fixtures (see Figures 1 and 2) and introduced into the production line for adhesive application and assembly in accordance with standard production methods. Frequency of test shall be at a minimum of one (1) every twenty-five (25) cartridges. After three (3) consecutive acceptable tests at this frequency, the frequency may be decreased to one (1) every fifty (50) cartridges. After three (3) consecutive acceptable tests at this frequency, the frequency may be decreased to one (1) every one hundred (100) cartridges. In the event of a failure, frequency shall revert back to one (1) every twenty-five (25) cartridges and all products produced since the last acceptable verification shall be rejected. Whenever the gluing of 120 mm combustible cartridge cases has been interrupted for a period of ninety (90) working days, the test frequency shall revert back to one (1) sample every twenty-five (25) cartridges.

4.4.2.1.1 Adhesive cure. - The test cartridge case shall be cured for two (2) hours minimum at ambient temperature prior to test.

4.4.2.1.2 Test of bond. - The test cartridge shall then be attached to a tensile test fixture and the required minimum load shall be applied at a maximum rate of 45 + 5 mm/minute with the test cartridge loaded to failure. Any loosening of the bond below 8900 N shall be considered a failure. Any failure of cartridge case material below 8900 N, other than at the glue joint, shall be recorded for information only, and the Government Contracting Officer shall be notified immediately.

4.4.2.2 Cartridge case coating. - The cartridge case coating requirement shall be verified by introducing a test vehicle into the cartridge coating operation and subjecting it to the same coating process as applied to cartridges in production. Frequency of the test shall be a minimum of one (1) every twenty-five (25) cartridges or one

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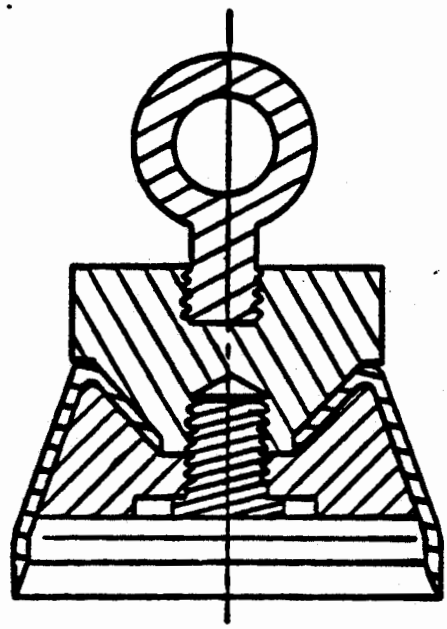


Figure 1. Case cover attachment fixture.

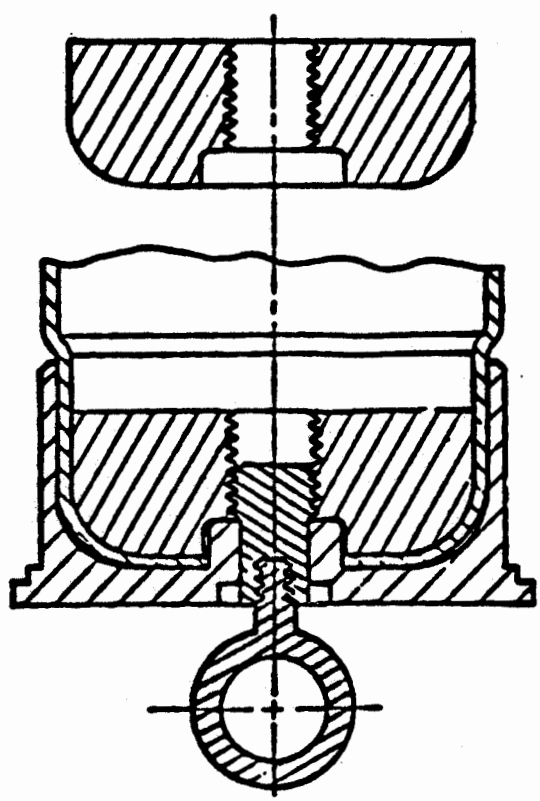


Figure 2. Case base attachment fixture.

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every hour whichever results in the greater number of samples. After three (3) consecutive acceptable checks have been made, frequency may decrease to one (1) every four hours. In the event of a failure, frequency shall revert back to one (1) every twenty-five (25) cartridges or one every hour and all product coated since the last acceptable verification shall be rejected. The first assembly shall be checked at the start of each shift and after a rejection.

4.4.2.2.1 Peeling. - The cartridge case shall be visually examined for evidence of peeling or delamination using samples prepared per paragraph 4.4.2.2. In production, peeling is cause for rejection of the cartridge.

4.4.2.3 Torque test. - The minimum assembly torque, as specified in the applicable drawing, shall be gradually applied to the applicable joint in the direction required for assembly. This test must be conducted while the adhesive/sealant is still wet. Any perceptible movement between the interfacing units shall be cause for rejection of all assemblies produced since the last acceptable check.

4.4.3 Projectile.

4.4.3.1 Body slug assembly.

4.4.3.1.1 Rubber material properties and molding. - See DOD-P-64009.

4.4.3.1.2 Nylon material properties and molding. - See DOD-P-64009.

4.4.4 Chamberability. - The cartridge shall be subjected to the chamber gage test as specified on the applicable cartridge drawing. Failure of the cartridge to completely enter the gage shall be cause for rejection of the cartridge.

4.4.5 Primer circuit electrical tests. - The electrical resistance of the cartridge/primer circuit shall be verified. Primer electrode to primer body resistance shall be measured prior to assembling into the cartridge. Transfer resistance between the primer body and case base shall be measured after the primer has been torqued into the case base but prior to staking it. Resistance shall be measured using instrumentation that will limit test current per drawing requirement. Failure to meet the specified requirement shall be cause for rejection of the assembly.

4.4.6 Performance tests. - Performance tests shall be conducted on test rounds, assembled from parts and assemblies which have been inspected in accordance with the applicable Quality Conformance Inspection tests specified in this document.

4.4.6.1 Test conditions.

4.4.6.1.1 Weapon. - Unless otherwise specified all performance tests shall be conducted using an M256 gun on a fixed mount.

4.4.6.1.2 Temperature. - Unless otherwise specified performance

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tests shall be conducted with ammunition conditioned at 63 °C - 3 °C for a period of 24 hours minimum prior to firing.

* 4.4.6.1.3 Cannon Proof Cartridge Preparation. - Tape protecting gage insertion hole shall be completely removed from the cartridge case prior to insertion of the round into the gun tube.

4.4.6.2 Test procedures.

4.4.6.2.1 Chamber pressure. - Chamber pressure shall be measured piezoelectrically at a location 8.9 cm from the rear face of the tube. The gage shall be positioned at 30° counterclockwise from the vertical center line.

4.4.6.2.2. Retest. - If requested by the Government contracting officer, an additional sample the same size as the original sample shall be provided for retest. All retests are at the discretion of the Government contracting officer.

4.4.6.3 Performance characteristics.

4.4.6.3.1 Chamber pressure. - Chamber pressure shall be verified piezoelectrically. Failure to meet the specified requirement shall be cause for rejection of the lot.

4.4.6.3.2 Misfire. - All firing tests shall be monitored for evidence of misfire. Any misfire shall be cause for rejection of the lot.

4.4.6.3.3 Metal parts security. - Metal parts security shall be verified during gun firing tests. Verification will be accomplished by placement of a down bore camera approximately 20 meters from muzzle. Any projectile that exhibits evidence of breakup or gas leak shall be cause for rejection of the lot.

4.4.6.3.4 Extraction. - The case base and seal assembly shall be extracted from the chamber after firing (See 6.5). Any case base and seal assembly that fails to extract shall be cause for rejection of the lot.

5.0 PACKAGING

5.1 Cleaning. - Not applicable.

5.2 Drying. - Not applicable.

5.3 Preservation. - Not applicable.

5.4 Unit and intermediate packaging.

5.4.1 Level A. - To be determined.

5.4.2 Level B. - To be determined.

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5.4.3 Level C. - The item shall be unit and intermediate packaged in accordance with paragraph 173.54 of the CFR 49 Parts 100-199. Additionally, appropriate moisture protection to the combustible cartridge case and propellant shall be provided.

5.5 Packing

5.5.1 Level A. - To be determined.

5.5.2 Level B. - To be determined.

5.5.3 Level C. - The package in paragraph 5.4.3 shall be packed in accordance with paragraph 173.54 of CFR 49 Parts 100-199.

5.6 Marking

5.6.1 Level A. - To be determined.

5.6.2 Level B. - To be determined.

5.6.3 Level C. - Marking to be in accordance with CFR 49 Parts 100-199 and MIL-STD-129.

5.7 Shipping. - When propellant from more than one lot is shipped at one time, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of lots in transit.

6.0 NOTES

6.1 Intended use. - The items covered by this specification are intended as test vehicles for use in the M256 cannon. The Cannon Proof Slug is for proof testing the M256 gun tube. With modified propellant loads, it is also used for the high temperature testing phase of the 120 mm case base and seal assemblies. The case base test slug is intended for use in the cold temperature testing of the 120 mm case base and seal assembly.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. - See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament Research and Development Center, Attn: AMSMC-QAT-I (D), Dover, NJ 07801.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or US Army Armament Research and Development Command drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Extraction. - The term extraction shall mean that the fired case base and seal assembly is cleared from the rear of the weapon by the automatic mechanism.

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6.6 Deterioration of primer. - If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to, and must satisfactorily pass, the check test for deterioration in accordance with MIL-P-70451 immediately before the primer lot is assembled into the cartridges. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the primer check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc, permitted while the primer lot was under the jurisdiction of either contractor.

6.7 Propellant weight determination.

6.7.1 Cannon proof slug cartridge charge establishment.

6.7.1.1 KE Slug JA2 propellant. - When KE slug JA2 propellant is used, the data from the CE/CV performed as part of lot acceptance in specification DOD-P-70639 shall be used to determine charge weight for that lot of propellant.

6.7.1.2 M829 JA2 propellant. - When M829 JA2 propellant is to be used, a CE/CV shall be performed on the identified lot per the procedure specified in specification DOD-P-70639, paragraph 4.5.8. The charge weight shall then be established using this data. If a lot fails to meet the requirements, it shall not be used in Cannon Proof cartridges.

6.7.2 Primer test cartridge charge weight. - The charge weight for primer test cartridges shall be such that the requirements of specification MIL-P-70451 are met.

6.7.3 Case base test cartridge charge weight. - The cartridge weight for case base test cartridges shall be such that the requirements of specification DOD-C-63485 are met.

6.8 Definitions of packaging levels.

6.8.1 Level A. - Full military protection for unrestricted (OCONUS) shipment and storage.

6.8.2 Level B. - Limited military protection for limited (OCONUS) shipment and storage.

6.8.3 Level C. - For interplant (CONUS) shipment and very limited storage.

6.8.4 For more detailed interpretation of the various military packaging levels, see AR-700-15.

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6.9 Keywords.
Ammunition
Cartridge
Cannon
Slug
Tank

Custodian:
Army-AR

Preparing activity:
Army-AR

Project Number:
1315-AD54

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